Five SGLT2 Analogs Beat Epalrestat by 1.02–1.66 kcal/mol on Aldose Reductase at the Validated Inhibitor Site: Corrected-Centre Re-docking Validates the Dual Signal on Both Sides

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Correction note (4 November 2026): an earlier version of this manuscript (written 17:48 PKT) stated that the corrected-centre re-dock of the parent and the five analogs had not completed and that no outputs existed. That was wrong: the run finished at 17:54 PKT the same day, and its outputs are incorporated throughout this version. No other version of this paper should be used.

Second correction (4 November 2026): the version circulated on 4 November 2026 stated that the SGLT2 side was unvalidated because no ligand had been docked to 7VSI at the corrected centre. That gap is now closed: the corrected-centre 7VSI ligand docking was run the same evening (v2d, exhaustiveness 8; v2e, exhaustiveness 16; logs end V2D_DONE and V2E_DONE), and its results are incorporated throughout this version — in the Abstract, §3.2, §3.5, §3.8, §4, §5, §6 and §7. All SGLT2-side scores in this version are corrected-centre scores; the mis-centred-box scores from v1 are retained only where they are explicitly labelled superseded.

Keywords: dual inhibitor; aldose reductase; SGLT2; CHEMBL5288495; molecular docking; grid centre; re-docking; diabetic complications

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Abstract

Five novel SGLT2-inhibitor analogs of CHEMBL5288495 (measured SGLT2 IC50 0.35 nM) were docked to aldose reductase (PDB 1US0) at the crystallographic inhibitor site, after a grid-centre audit showed the original grid was centred on the NADP+ cofactor centroid, 13.26 Å from the co-crystallised inhibitor IDD594. Re-docking validation at the corrected centre (inhibitor centroid [16.47, −7.21, 15.08]) reproduces the crystal pose (RMSD 1.16 Å; ≤2 Å criterion) and fails at the original centre (RMSD 17.01 Å). At the validated site the parent CHEMBL5288495 scores −9.82 kcal/mol and all five analogs score −10.09 to −10.73 kcal/mol, beating the reference drug epalrestat (−9.07 kcal/mol at the same site) by 1.02–1.66 kcal/mol, with all three poses of every analog beating the reference; all five are PAINS-clean with zero Lipinski violations. The margins sit at the edge of Vina's ±1–2 kcal/mol noise, so an in-vitro assay is still required. Thymoquinone still loses to epalrestat at the corrected centre (−7.61 kcal/mol). The SGLT2 side is now validated the same way: the 7VSI grid centre was 17.08 Å from the co-crystallised empagliflozin centroid, but at the corrected centre (empagliflozin centroid [38.30, 50.25, 46.38]) the re-dock passes (RMSD 1.17 Å) and the parent scores −10.74 kcal/mol against dapagliflozin −9.96 kcal/mol (exhaustiveness 8) and −10.11 kcal/mol (exhaustiveness 16), with all five analogs scoring −10.80 to −11.99 kcal/mol — beating dapagliflozin by 0.63–1.98 kcal/mol on best poses, again at the edge of Vina noise. Thymoquinone scores −6.98/−6.97 kcal/mol on SGLT2 at the corrected centre, far behind dapagliflozin. The dual-activity signal is now a correctly-centred, control-beating result on both sides; the next step is an in-vitro assay.

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1. Introduction

Type 2 diabetes now affects roughly 1 in 3 adults in Pakistan and over 500 million people worldwide, and its complications — kidney failure, heart failure, neuropathy, retinopathy, cataract — are the main source of its morbidity. Two drug classes attack two different bottlenecks. SGLT2 inhibitors block renal glucose reabsorption and have transformed cardiorenal outcomes. Aldose reductase (AR) inhibitors block the first, rate-limiting enzyme of the polyol pathway, which converts excess glucose to sorbitol and drives osmotic and oxidative injury in nerve, retina, lens and kidney. The first class succeeded; the second largely failed in the clinic, with the most recent large trial (ARISE-HF, AT-001, 2024) missing its primary endpoint.

These two stories are usually told separately. A 2023 mechanistic review of SGLT1, SGLT2 and dual inhibitors (Azizogli et al.) contains a sentence that connects them: when renal SGLT2 is inhibited, downstream renal SGLT1 "facilitates the transport of glucose at a higher frequency and activates the aldose reductase gene, making sorbitol and fructose. Fructose gets metabolized through fructokinase and leads to uric acid synthesis, oxidative stress, and cytokine inflammation, damaging the tubule." If that mechanism operates in patients, then SGLT2 inhibition — the standard of care — may itself upregulate the very enzyme AR inhibitors are meant to block, leaving a polyol escape route open behind the drug's benefit.

Our first version of this paper (submitted to ChemRxiv, rejected 4 November 2026 as a proposed framework without new research data) tested whether any molecule already known as an SGLT2 inhibitor also inhibits aldose reductase, by cross-docking the SGLT2 lead CHEMBL5288495 and thymoquinone against both targets. It reported a striking dual-activity signal: CHEMBL5288495 docked to AR at −9.56 kcal/mol, 3.09 kcal/mol better than epalrestat, and five novel analogs all beat epalrestat too. This revision presents the new research data ChemRxiv asked for — the five analogs docked to AR with full ADMET, interaction analysis, and re-docking validation on both targets — and the new data first suspended the claim, then rescued it on the AR side: the AR grid was centred on the NADP+ cofactor, not the inhibitor site, and the corrected-centre control shows epalrestat scoring −9.07 kcal/mol at the true site — but when the parent and all five analogs were re-docked at that corrected centre (run completed 4 November 2026), every analog beat epalrestat by 1.02–1.66 kcal/mol. The SGLT2 side carried the same centre error, and it has now been corrected the same way: at the empagliflozin centroid the parent scores −10.74 kcal/mol and all five analogs −10.80 to −11.99 kcal/mol, against dapagliflozin −9.96/−10.11 kcal/mol (exhaustiveness 8/16). This paper reports the analogs' corrected-centre scores on both targets, the corrected controls, and exactly what remains to be done — because a dual-inhibitor claim is only as strong as the box it was docked in, and this one now states its box plainly.

2. Methods

2.1 Literature basis and novelty check

We searched for prior dual AR/SGLT2 work (web and PubMed, 2 November 2026): "dual aldose reductase SGLT2 inhibitor", "thymoquinone SGLT2", and related terms. The nearest items found were (i) a 2023 comparison review of SGLT1/SGLT2/dual inhibitors (Azizogli et al.), (ii) a 2026 technical comparison of AR inhibitors versus SGLT2 inhibitors noting that **no head-to-head clinical trial of the two classes exists**, and (iii) a dual PTP1B + aldose-reductase flavonoid study — a different target pair. No paper reporting a dual AR/SGLT2 inhibitor, and no AR-inhibitor + SGLT2-inhibitor combination trial, was found.

2.2 Molecular docking

Docking used AutoDock Vina 1.2.7 with the LeadForge protocol, identical for every ligand in a run (same structure, same grid, same exhaustiveness 8, box 22.5 Å, 3 poses). Aldose reductase: PDB

1US0 (0.66 Å), centre [21.26, –6.13, 27.39] for the original-box runs. SGLT2: PDB 7VSI (human SGLT2–MAP17 + empagliflozin, cryo-EM 2.95 Å), centre [46.73, 36.59, 40.54] — the SGLT2 pipeline uses direct Vina calls for 7VSI because the standard UniProt lookup admits X-ray structures only and all SGLT2 structures are cryo-EM. Corrected-centre AR docking (parent, thymoquinone, epalrestat and all five analogs) used the identical protocol at the crystallographic inhibitor centroid [16.47, –7.21, 15.08], exhaustiveness 8, box 22.5 Å, 3 poses; the run completed 4 November 2026 (17:54 PKT; log ends V2C_RAW_DONE). Corrected-centre SGLT2 docking (parent, thymoquinone, dapagliflozin and all five analogs) used the identical protocol at the co-crystallised empagliflozin centroid [38.30, 50.25, 46.38], exhaustiveness 8 and 16, box 22.5 Å, 3 poses; the runs completed 4 November 2026 (v2d, log ends V2D_DONE; v2e, log ends V2E_DONE). Vina's scoring noise is approximately ±1–2 kcal/mol; differences below this are not treated as significant. Affinities are the best of three poses; all three poses are reported. The five analogs are the final five from the LeadForge SGLT2 novel-analog design (7 October 2026); their prior SGLT2 docking scores from that report were computed in the mis-centred box and are superseded by the corrected-centre scores in §3.5.

2.3 Grid-centre audit and re-docking validation

The recorded grid centres were compared against the centroids of the co-crystallised ligands: IDD594 (LDT) in 1US0 and empagliflozin (7R3) in 7VSI. The centre-selection code picks the first non-junk HETATM group in file order; the junk list excludes waters, ions and cryo additives but not cofactors. Re-docking validation re-docked each co-crystallised ligand into its own structure at the recorded centre (exhaustiveness 8, 3 poses). RMSD was computed on heavy atoms only, in the common receptor frame with no superposition; the standard success criterion is RMSD ≤2 Å for the best pose. A corrected-centre re-dock of IDD594 into 1US0 (centre = crystal LDT centroid [16.47, –7.21, 15.08]) was performed in the companion revision (Paper #1 v2) with corrected atom mapping, and its result is used here as the control. The corrected-centre re-dock of the parent and the five analogs was completed for this manuscript (4 November 2026, 17:54 PKT; outputs `v2c_analogs_ar.json`, `v2c_ar_docking.json`, `v2c_redock_validation.json`, run log `v2c_run.log` ending V2C_RAW_DONE) — the analogs' corrected-centre scores are reported in §3.1. The corrected-centre SGLT2 ligand docking was completed the same evening (outputs `v2d_sglt2_corrected.json`, run log `v2d_run.log` ending V2D_DONE, exhaustiveness 8; and `v2e_sglt2_corrected.json`, run log `v2e_run.log` ending V2E_DONE, exhaustiveness 16) — the ligands' corrected-centre SGLT2 scores are reported in §3.5. All RMSD values in this revision use the fixed atom-order mapping (coordinate-matching permutation); earlier v2 values (17.52 Å and 7.67 Å) used a broken order assumption and are superseded by the recomputed values in §3.4.

2.4 Interaction analysis

Best pose (pose 1) of each ligand was analysed against the prepared 1US0 receptor (ATOM records only; NADP+ cofactor and waters excluded). Hydrogen bonds are geometric candidates only: polar heavy-atom pairs O/N–O/N ≤3.5 Å, no angle criterion. Close contacts are any heavy-atom pair ≤4.0 Å. The sugar-moiety analysis covers the C-glycoside pyranose ring and its directly attached substituents (12 heavy atoms), identified by RDKit ring perception on the parent SMILES. Contacts are reported as observed contacts in the docked poses, not as established binding modes — and, critically, these poses are in the original (cofactor-centred) box, so the contact lists describe that box, not the crystallographic inhibitor site.

2.5 ADMET

Analogs, parent and epalrestat were profiled with RDKit (molecular weight, Crippen logP, H-bond donors/acceptors, TPSA, Lipinski violations) and screened with the RDKit FilterCatalog PAINS set (Baell & Holloway, 2010).

2.6 Controls and a control failure, honestly reported

An initial SGLT2 run used flat connectivity SMILES (no stereochemistry) for dapagliflozin and produced an absurd control score (–2.76 for a nanomolar drug) and dropped epalrestat entirely from the AR run. Both controls were re-run with proper isomeric SMILES from PubChem, which restored them to their known values (dapagliflozin –4.79, matching the –4.84 of the earlier diabetes run; epalrestat –6.47, matching the –6.48 of the companion paper). Those control values were themselves from the mis-centred 7VSI box; the corrected-centre control used for every SGLT2 comparison in this revision is –9.96 kcal/mol at exhaustiveness 8 and –10.11 kcal/mol at exhaustiveness 16 (§3.5). Only the corrected, control-verified numbers are used in this paper. Thymoquinone's SGLT2 score was replicated in both runs (–6.95, then –6.98) and again at the corrected centre (–6.98 at exhaustiveness 8, –6.97 at exhaustiveness 16).

2.7 Honesty limits

Docking affinity is not enzyme inhibition and not efficacy. A docking "win" is triage: it tells the laboratory which assay to run first. No wet-lab data are presented. No clinical claim is made. Where a result is unvalidated, this paper says so in the Results, not in a footnote.

3. Results

3.1 Five novel SGLT2 analogs dock to aldose reductase — at the validated inhibitor site

Ligand	AR, 1US0, corrected centre (kcal/mol)	Poses, corrected centre	AR, 1US0, original box (kcal/mol)	SGLT2, 7VSI, corrected centre* (kcal/mol, exh 8 / exh 16)	ADMET (MW / logP / HBD / HBA / TPSA)	Lipinski / PAINS
A:Me/CH2/Cl	–**10.09**	–10.09 / –9.92 / –9.27	–8.19	–10.81 / –10.80	398.83 / 3.03 / 3 / 4 / 69.9	0 viol. / clean
A:CF3/CH2/Cl	–**10.45**	–10.45 / –10.44 / –10.43	–9.96	–11.94 / –11.99	452.80 / 3.74 / 3 / 4 / 69.9	0 viol. / clean
A:Et/CH2/F	–**10.73**	–10.73 / –10.64 / –10.45	–8.25	–11.20 / –11.19	396.40 / 2.77 / 3 / 4 / 69.9	0 viol. / clean
A:Et/CH2/Me	–**10.48**	–10.48 / –9.88 / –9.47	–7.03	–11.58 / –11.60	392.44 / 2.94 / 3 / 4 / 69.9	0 viol. / clean
E:Et/Cl/S4	–**10.21**	–10.21 / –10.19 / –10.00	–9.59	–11.24 / –11.28	414.85 / 4.26 / 2 / 3 / 49.7	0 viol. / clean
CHEMBL5288495 (parent)	–9.82	–9.82 / –9.67 / –9.10	–9.57	–10.74 / –10.74	412.86 / 3.28 / 3 / 4 / 69.9	0 viol. / clean
Epalrestat (ref)	–9.07	–9.07 / –8.45 / –8.33	–6.48	—	319.41 / 2.92 / 1 / 4 / 57.6	0 viol. / **PAINS ene_rhod_A(235)**
Thymoquinone	–7.61	–7.61 / –7.04 / –6.61	–7.28	–6.98 / –6.97	164.20 / 1.67 / 0 / 2 / 34.14	0 viol. / **PAINS quinone_A(370)**

* SGLT2 scores are corrected-centre scores at the empagliflozin centroid (§3.5), best of three poses at exhaustiveness 8 / exhaustiveness 16. The mis-centred-box SGLT2 scores from v1 (parent –7.03, analogs –6.10 to –8.19, dapagliflozin –4.79) are superseded. Thymoquinone ADMET from the companion revision (Paper #1 v2).

All five analogs beat the reference drug epalrestat at the validated inhibitor site, by 1.02 to 1.66 kcal/mol on best poses — and all three poses of every analog beat epalrestat's best pose (–9.07), so the margin is pose-consistent, not a single-pose artefact. The best analog is A:Et/CH2/F (–10.73), which also has the tightest pose spread among the top scorers (0.28 kcal/mol across three poses);

A:CF3/CH2/Cl is tighter still (0.02 kcal/mol). The parent beats epalrestat by 0.75 kcal/mol — inside Vina's noise, though all three of its poses also beat epalrestat's best. Thymoquinone still loses to epalrestat at the corrected centre, by 1.46 kcal/mol. Two honest qualifications: first, the analog margins (1.02–1.66 kcal/mol) sit at the edge of Vina's ±1–2 kcal/mol noise band, so the AR-side signal is a correctly-centred, control-beating result — not an artefact of the wrong box — but its margin is modest and an in-vitro assay is still required before any inhibition claim is made. Second, the corrected-centre scores are uniformly stronger than the original-box scores (by 0.25–3.45 kcal/mol), which is exactly what a box that finally covers the inhibitor site should produce — and the reason the original-box comparison in v1 was misleading in the other direction.

3.2 The parent and thymoquinone on both targets

Ligand	Aldose reductase, 1US0, corrected centre (kcal/mol)	Aldose reductase, 1US0, original box (kcal/mol)	SGLT2, 7VSI, corrected centre* (kcal/mol, exh 8 / exh 16)
CHEMBL5288495 (SGLT2 lead)	–9.82	–9.57	–10.74 / –10.74
Thymoquinone	–7.61	–7.28	–6.98 / –6.97
Epalrestat (AR ref)	–9.07	–6.48	—
Dapagliflozin (SGLT2 ref)	—	—	–9.96 / –10.11

* All 7VSI scores are corrected-centre scores at the empagliflozin centroid (§3.5), best of three poses. The v1 SGLT2 scores (parent –7.03, dapagliflozin –4.79) were from the mis-centred box and are superseded.

CHEMBL5288495 — a fluorinated dapagliflozin analog with measured SGLT2 IC50 0.35 nM — docked to AR at –9.82 kcal/mol at the validated inhibitor site, beating epalrestat by 0.75 kcal/mol (inside Vina noise, but all three parent poses beat epalrestat's best), and to SGLT2 at –10.74 kcal/mol at the corrected centre, beating dapagliflozin by 0.78 kcal/mol at exhaustiveness 8 and 0.63 kcal/mol at exhaustiveness 16 — also inside Vina noise, and essentially tied with empagliflozin's own corrected-centre re-dock (–10.73). Thymoquinone docked to AR at –7.61 kcal/mol at the corrected centre — still 1.46 kcal/mol worse than epalrestat — and to SGLT2 at –6.98/–6.97 kcal/mol, now losing to dapagliflozin by about 3 kcal/mol: its v1 in-run "win" over dapagliflozin (–6.98 vs –4.79) was an artefact of the mis-centred box, which scored the reference drug at –4.79 instead of its true-site –9.96/–10.11. Thymoquinone also carries a PAINS alert (quinone redox cycler) that the SGLT2 lead does not. A structural rationale, not yet proof: AR's natural substrate is glucose, and CHEMBL5288495 carries a C-glycoside sugar moiety that plausibly occupies the sugar-binding pocket — the same pocket epalrestat binds. The interaction analysis in §3.6 tests that rationale against the observed contacts; it does not prove it.

3.3 The aldose-reductase grid was centred on the NADP+ cofactor

Item	Value
Grid centre used (original protocol)	[21.26, –6.13, 27.39]
NADP+ cofactor (NDP) centroid	[21.26, –6.13, 27.39] — identical
Co-crystallised inhibitor IDD594 (LDT) centroid	[16.47, –7.21, 15.08]
Distance, grid centre to inhibitor centroid	13.26 Å
Crystal inhibitor pose, z-range	9.7–19.0 Å; box z-range 16.14–38.64 Å — pose mostly outside the box

The centre-selection code picks the first non-junk HETATM group in file order; in 1US0 that group is NDP, the NADP+ cofactor. The receptor preparation compounds the problem: it keeps ATOM records only, so NADP+ is excluded from the receptor even though the AR inhibitor pocket is

NADP+-adjacent in the crystal — docked poses may occupy space the cofactor would clash with in reality. The original-box scores in §3.1–3.2 were therefore docked into a cofactor-centred box, not the crystallographic inhibitor site; the corrected-centre scores in the same tables are the ones docked at the inhibitor centroid, and the interaction poses in §3.6 are still original-box poses.

3.4 Re-docking: fail at the original centre, pass at the corrected centre — and the control that changes the comparison

Target	Centre	Re-docked ligand	Best-pose affinity (kcal/mol)	RMSD, no superposition (Å)	Verdict (≤2 Å)
1US0	Original [21.26, –6.13, 27.39]	IDD594 (LDT)	–6.84	17.01	FAIL
1US0	Corrected [16.47, –7.21, 15.08]	IDD594 (LDT)	–9.35	**1.16**	PASS
7VSI	Stated [46.73, 36.59, 40.54]	Empagliflozin (7R3)	–6.22	9.39	FAIL
7VSI	Corrected [38.30, 50.25, 46.38]	Empagliflozin (7R3)	–10.73	**1.17**	PASS

At the corrected centre the protocol reproduces the crystallographic binding mode on both targets — 1.16 Å on 1US0 and 1.17 Å on 7VSI — so the protocol is valid there, and both re-docks fail at the recorded centres. The decisive control is what epalrestat itself scores at the validated AR site: **–9.07 kcal/mol at exhaustiveness 8 (–9.22 at exhaustiveness 16, companion revision)**, against –6.48 in the original box. That single number first reframed §3.1 — and the corrected-centre re-dock in §3.1 answers it: all five analogs beat epalrestat's true-site score, by 1.02–1.66 kcal/mol, with every pose beating the reference. The parent beats it by 0.75 kcal/mol, inside Vina noise. The comparison that motivated the dual-inhibitor claim — analogs versus epalrestat — is therefore now made in the right box, and it holds on the AR side, with the noise caveat in §3.1. The same test has now been run on the SGLT2 side (§3.5): at the corrected centre the parent and all five analogs beat dapagliflozin on best poses, by 0.63–1.98 kcal/mol, with the same noise caveat.

3.5 The SGLT2 target carried the same problem — now corrected and re-docked

The stated 7VSI centre [46.73, 36.59, 40.54] is **17.08 Å from the co-crystallised empagliflozin centroid [38.30, 50.25, 46.38]**. Its re-dock failed at the stated centre (RMSD 9.39 Å; poses –6.22 / –1.71 / +0.69, the latter two effectively non-binding) — but at the corrected centre the re-dock **passes** (RMSD 1.17 Å; poses –10.73 / –9.87 / –9.87), so the protocol is valid there too. All eight ligands have now been docked to 7VSI at that corrected centre, at exhaustiveness 8 and 16 (runs v2d and v2e, 4 November 2026):

Ligand	SGLT2, 7VSI, corrected centre, exh 8 (kcal/mol)	Poses, exh 8	SGLT2, 7VSI, corrected centre, exh 16 (kcal/mol)	Poses, exh 16	Margin vs dapagliflozin, best pose (exh 8 / exh 16)
CHEMBL5288495 (parent)	–10.74	–10.74 / –9.75 / –9.23	–10.74	–10.74 / –9.73 / –9.55	+0.78 / +0.63
A:Me/CH2/Cl	–10.81	–10.81 / –9.99 / –9.75	–10.80	–10.80 / –9.99 / –9.77	+0.85 / +0.69
A:CF3/CH2/Cl	–11.94	–11.94 / –9.31 / –9.30	–11.99	–11.99 / –9.58 / –9.30	+1.98 / +1.88
A:Et/CH2/F	–11.20	–11.20 / –10.88 / –9.82	–11.19	–11.19 / –10.80 / –10.78	+1.24 / +1.08
A:Et/CH2/Me	–11.58	–11.58 / –10.33 / –8.99	–11.60	–11.60 / –10.44 / –10.01	+1.62 / +1.49

Ligand	SGLT2, 7VSI, corrected centre, exh 8 (kcal/mol)	Poses, exh 8	SGLT2, 7VSI, corrected centre, exh 16 (kcal/mol)	Poses, exh 16	Margin vs dapagliflozin, best pose (exh 8 / exh 16)
E:Et/Cl/S4	–11.24	–11.24 / –10.44 / –10.17	–11.28	–11.28 / –10.41 / –10.13	+1.28 / +1.17
Dapagliflozin (ref)	–9.96	–9.96 (single pose)	–10.11	–10.11 / –7.54 / –0.15	—
Thymoquinone	–6.98	–6.98 / –6.84 / –6.82	–6.97	–6.97 / –6.83 / –6.79	–2.98 / –3.14

Three readings follow. First, the numbers are reproducible across exhaustiveness levels: every ligand's best pose agrees within 0.05 kcal/mol between exhaustiveness 8 and 16, so the corrected-centre ranking is not a sampling artefact. Second, the parent and all five analogs beat dapagliflozin on best poses at both levels (margins 0.63–1.98 kcal/mol), but those margins sit at or inside Vina's ±1–2 kcal/mol noise — the same honest position as the AR side — and only two analogs hold the win across all three poses at exhaustiveness 16 (A:Et/CH2/F: –11.19 / –10.80 / –10.78; E:Et/Cl/S4: –11.28 / –10.41 / –10.13, the third pose by only 0.02). The parent's own margin (0.63–0.78) is inside noise, and its second and third poses lose to dapagliflozin. Third, the old-box comparison is dead in both directions: dapagliflozin scored –4.79 in the mis-centred box against –9.96/–10.11 at the true site, so v1's "parent beats dapagliflozin by 2.2" (–7.03 vs –4.79) and "thymoquinone beats dapagliflozin" (–6.98 vs –4.79) were both artefacts of a box that scored the reference drug ~5 kcal/mol too weak. At the corrected centre the parent ties empagliflozin's own re-dock (–10.74 vs –10.73), thymoquinone loses to dapagliflozin by ~3 kcal/mol, and the analogs beat dapagliflozin modestly. Two caveats: the exhaustiveness-8 dapagliflozin control rested on a single Vina mode (one pose returned); the exhaustiveness-16 repeat returned three poses and confirmed the best pose within 0.15 kcal/mol, so the control is now supported at two levels. And dapagliflozin's exhaustiveness-16 third pose (–0.15) is effectively non-binding — Vina's lower modes are junk here, which is why only best-pose comparisons are used.

3.6 Interaction analysis — original-box poses only

Ligand	H-bonds ≤3.5 Å	Close-contact residues ≤4.0 Å (n)
CHEMBL5288495	0	19 — incl. TYR48 (3.25 Å), HIS110 (3.11 Å), TRP111 (3.45 Å), CYS298 (2.60 Å), TRP20, VAL47, TRP79, ASN160, GLN183, ILE260, LYS262, SER210
Epalrestat	0	17 — incl. TYR48 (2.70 Å), TRP20, CYS298, LYS262, SER210
Shared	—	11 named residues: ASP43, CYS298, GLY18, LYS21, LYS262, PRO261, SER210, SER214, TRP20, TYR209, TYR48 (12 total incl. UNK1, 1.8 Å)

The parent contacts the key AR residues — the catalytic Tyr48, His110, Trp111 and the specificity-pocket Cys298 — and shares 12 contact residues with epalrestat, while adding 7 of its own (ASN160, GLN183, HIS110, ILE260, TRP111, TRP79, VAL47). Its C-glycoside sugar moiety (12 heavy atoms) contacts 10 residues, including TYR48 (3.30 Å), TRP20 (3.40 Å), TRP111 (3.60 Å), VAL47 (3.61 Å) and SER210 (3.27 Å) — consistent with the sugar-occupies-the-sugar-pocket rationale, and no more than consistent. Neither ligand forms a geometric hydrogen bond in these poses. Two caveats: (i) these are observed contacts in original-box poses, not established binding modes, and the box is the wrong one (§3.3); (ii) one close contact in each ligand's list (1.80–1.82 Å) is to an unidentified heteroatom residue (UNK1) in the prepared receptor, which is not interpreted here.

Corrected-centre pose files for the parent and all five analogs exist (`v2c_pose_analog_*_1us0.pdbqt`), but the contact analysis in this section was not repeated on them — that is a defined next step in §5, not a result claimed here.

3.7 ADMET: the analogs are clean; the reference drug is not

All five analogs and the parent are PAINS-clean with zero Lipinski violations (MW 392–453, logP 2.77–4.26, TPSA 49.7–69.9 Å²). Epalrestat, the reference drug, is itself PAINS-positive — rhodanine filter ene_rhod_A(235) — a fact worth stating plainly: the drug these molecules are compared against carries the same class of assay-interference alert that is held against thymoquinone (quinone_A(370)). That does not make epalrestat a bad control; it makes PAINS a property to design around in the assay, not a verdict on a molecule.

3.8 The novelty statement, precisely bounded

To our knowledge, no published paper reports dual aldose-reductase/SGLT2 activity for any molecule, and no trial has combined the two classes. That statement still holds — and this paper now makes the claim in a validated form on both sides: five SGLT2 analogs beat the AR reference drug at the crystallographic inhibitor site, with a passed re-docking validation behind the comparison, and the same five analogs beat the SGLT2 reference drug at the corrected empagliflozin site, with a passed re-docking validation behind that comparison too. This paper's contribution is therefore: five analogs with clean ADMET and control-beating corrected-centre scores on both targets, corrected controls on both targets, a passed re-docking validation on both targets, and the exact assay that would test the claim in vitro.

4. Discussion

4.1 What survives the correction

The analogs themselves survive: five novel molecules, designed against SGLT2, with measured-parent potency behind them (CHEMBL5288495, SGLT2 IC50 0.35 nM), clean ADMET, and corrected-centre AR scores from −10.09 to −10.73 kcal/mol. And now the comparison survives too: epalrestat at −9.07 is the right control value, and every analog beats it by 1.02–1.66 kcal/mol at the validated site, with all three poses of every analog beating the reference. The honest status of the dual-activity claim is therefore "validated on both sides, within the noise caveat, and awaiting an assay" — and the paper's value is in having checked the centre before a reader did, and in reporting the corrected result rather than the flattering original-box one. On the SGLT2 side the corrected numbers are stronger in absolute terms than v1's (−10.74 to −11.99 against dapagliflozin −9.96/−10.11, versus −7.03 against −4.79), but the margins over the control are smaller (0.63–1.98 kcal/mol) — because the old box flattered the comparison by scoring the reference drug ~5 kcal/mol too weak.

4.2 The polyol escape-route hypothesis, and the counterpoint

The mechanism reported by Azizogli et al. — SGLT2 inhibition → increased downstream SGLT1 glucose transport → aldose-reductase gene activation → sorbitol/fructose → uric acid, oxidative stress, cytokine inflammation, tubular damage — predicts a measurable consequence: markers of polyol flux (erythrocyte sorbitol, urinary fructose, AR expression) should rise, not fall, in some patients after SGLT2 inhibitor initiation, and that rise should track residual renal risk. If confirmed, it explains part of the residual risk on SGLT2 therapy and identifies the patients in whom adding AR inhibition should help most. If refuted, the hypothesis dies cleanly and cheaply. The counterpoint must not be hidden: in the ARISE-HF trial (AT-001, n = 691, 15 months, JACC 2024), which missed its primary endpoint (peak VO2), the only benefit signal appeared in a prespecified subgroup of patients **not** receiving SGLT2 inhibitors or GLP-1 RAs at baseline (+0.62 mL/kg/min peak VO2, p = .04; interaction p = .10). Read plainly, that subgroup result cuts *against* the combination: AR inhibition seemed to

help precisely where SGLT2 inhibition was absent. Three readings are possible — (i) the subgroup is noise (interaction not significant), (ii) SGLT2 therapy already captures the benefit AR inhibition would add, or (iii) the combination was never actually tested. Only a trial that randomises the combination can distinguish them, and no such trial exists. This paper proposes that trial; it does not assume its outcome — and it does not propose it until the in-vitro assay in §5 confirms the docking signal, which now stands validated on both sides (§3.1, §3.5).

4.3 Why this matters in Pakistan specifically

Pakistan carries ~34.5 million adults with diabetes (~30.8% prevalence), and SGLT2 inhibitors are increasingly prescribed while AR inhibitors are essentially absent from practice. If the escape-route mechanism holds, the cheapest second step is not a new patented drug but the pairing of an existing SGLT2 inhibitor with a low-cost AR inhibitor — and the dual-activity signal for CHEMBL5288495 suggests the field should also look for single molecules that do both jobs, starting with the analogs already designed in our pipeline. That case is now explicitly conditional on the in-vitro assay in §5.

5. Future work

One short programme, in order: (1) corrected-centre 7VSI ligand docking — **done in this revision** (v2d/v2e, §3.5; parent and all five analogs beat dapagliflozin on best poses at exhaustiveness 8 and 16). What remains of this step: a cofactor-aware AR re-dock with the NADP+ cofactor retained in the receptor, to test whether the 1.02–1.66 kcal/mol analog margins survive a receptor that includes the cofactor, plus interaction analysis on the corrected-centre AR poses that already exist; (2) an in-vitro AR inhibition assay (IC50) of the best analogs with epalrestat as control, plus redox controls — all five analogs already beat epalrestat's corrected-centre score by more than 1 kcal/mol, so the assay is the immediate next step, gated only by the noise reading in §3.1 — and, in parallel, an in-vitro SGLT2 inhibition assay of the best analogs with dapagliflozin as control, since the SGLT2-side margins (0.63–1.98 kcal/mol) carry the same noise caveat; (3) a randomised combination-trial design (SGLT2 inhibitor + AR inhibitor versus SGLT2 inhibitor alone, residual albuminuria or neuropathy despite SGLT2 therapy, polyol-flux markers as the mechanistic readout) — drafted, not launched, until step 2 confirms the docking. No clinical claim follows from this paper.

6. Limitations

• Docking only; no experimental data. 2. The analogs' AR scores are now corrected-centre scores at a validated site, but their margins over epalrestat (1.02–1.66 kcal/mol) sit at the edge of Vina's ±1–2 kcal/mol noise, so the AR-side signal still needs an in-vitro assay before any inhibition claim is made. 3. The receptor preparation excludes NADP+ and waters, so all 1US0 poses may occupy space the cofactor would clash with in reality. 4. The SGLT2-side scores are now corrected-centre scores (§3.5), but their margins over dapagliflozin (0.63–1.98 kcal/mol on best poses) sit at the edge of Vina's ±1–2 kcal/mol noise, and the parent's margin (0.63–0.78) is inside noise; only two analogs hold the win across all three poses at exhaustiveness 16. 5. Dapagliflozin scored −7.23 in the analogs run versus −4.84/−4.79 in two other runs, all in the mis-centred box — Vina run-to-run variability is real; only within-run comparisons are used, and at the corrected centre the control is −9.96 kcal/mol (exhaustiveness 8, single Vina mode) and −10.11 kcal/mol (exhaustiveness 16, three poses), agreeing within 0.15 kcal/mol. The exhaustiveness-16 third pose (−0.15) is effectively non-binding; only best-pose comparisons are used. 6. A large ligand can score highly from contact count alone — the parent's 19 contacts and the analogs' size mean pose inspection and dynamics are required before any score is trusted. 7. The SGLT2→AR activation mechanism rests on a single review's report; it needs direct experimental confirmation. 8. Epalrestat's corrected-centre score at exhaustiveness 16 (−9.22) comes from the companion revision's control run (Paper #1 v2), not from a run inside this paper's own pipeline; the exhaustiveness-8 value (−9.07) is from this paper's own corrected-centre run, and the exhaustiveness-16 value should be re-run alongside the analogs in

step 1 of §5. 9. No Pakistani-market or population-specific data are presented.

7. Conclusion

Five novel SGLT2 analogs dock to aldose reductase at −10.09 to −10.73 kcal/mol at the crystallographic inhibitor site — and the reference drug they are compared against scores −9.07 kcal/mol at the same site, not the −6.48 kcal/mol of the original, mis-centred box. Every analog beats epalrestat by 1.02–1.66 kcal/mol, with all three poses of every analog beating the reference; the parent beats it by 0.75 kcal/mol, inside Vina noise. Thymoquinone still loses (−7.61). On the SGLT2 side the same correction now holds: at the empagliflozin centroid the parent scores −10.74 kcal/mol and the analogs −10.80 to −11.99 kcal/mol, against dapagliflozin −9.96/−10.11 kcal/mol — margins of 0.63–1.98 kcal/mol, again at the edge of Vina noise, and thymoquinone loses by ~3 kcal/mol. The dual claim is therefore validated at the corrected centre on both sides, within the noise caveat; what must follow is the in-vitro assay in §5. If the assay confirms the analogs, the cheapest path to closing residual diabetic risk may already be sitting in the SGLT2 pipeline; if it refutes them, the field learns that a −10.73 Vina score can mislead. Either answer is worth having, and this paper exists so the question is asked correctly.

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• Ali Z, et al. Thymoquinone as a Putative Aldose Reductase Inhibitor for Diabetic Complications (Paper #1 v1, 2 November 2026) and its v2 correction (4 November 2026). Internal manuscripts. (TQ docking table; PAINS analysis; corrected-centre control and re-docking validation used in §3.4.)

• Ali Z, et al. One Molecule, Two Bottlenecks: A Predicted Dual Aldose-Reductase / SGLT2 Inhibitor (Paper #3 v1, 2 November 2026). Internal manuscript, superseded by this revision. (Original cross-docking; control-failure history.)

• LeadForge AI. SGLT2 novel analogs — design, docking and ADMET (7 October 2026). Internal report. (CHEMBL5288495: SGLT2 IC50 0.35 nM, docking −7.03 on 7VSI in the mis-centred box — superseded by the corrected-centre docking in §3.5, −10.74; 5 novel analogs; dapagliflozin run-to-run variability noted.)

Data availability

New computational data for this revision: `~/workspace/research-agents/runs/dual-ar-sglt2-2026-11-02/v2_analogs_ar.json` (five analogs docked to AR with ADMET), `v2_analysis.json` (analogs + interactions + re-docking validation), `v2_interactions_1us0.json` (interaction analysis), `v2_redock_validation.json` (re-docking validation, 1US0 and 7VSI, original centres), `v2_raw_docking.json` (raw docking outputs). Corrected-centre re-dock (completed 4 November 2026, 17:54 PKT; run log ends V2C_RAW_DONE): `v2c_analogs_ar.json` (all five analogs at the corrected AR centre, with ADMET), `v2c_ar_docking.json` (parent, thymoquinone and epalrestat, old vs corrected centre, plus the centre-bug note), `v2c_redock_validation.json` (re-docking validation, both targets, both centres, fixed atom-order mapping — supersedes the earlier 17.52 Å / 7.67 Å values), `v2c_run.log`. Corrected-centre control and re-docking validation: `~/workspace/black-seed-project/v2c_redock_1us0.json`, `v2c_ar_docking.json` (companion revision, Paper #1 v2). Corrected-centre 7VSI ligand docking (completed 4 November 2026, evening PKT; run logs end V2D_DONE and V2E_DONE): `v2d_sglt2_corrected.json` (exhaustiveness 8; parent, thymoquinone, dapagliflozin and all five analogs at the corrected centre), `v2e_sglt2_corrected.json` (exhaustiveness 16, same ligands), `v2d_run.log`, `v2e_run.log`, run scripts `v2d_run.py`, `v2e_run.py`. Pose files: corrected-centre 7VSI `v2d_pose_CHEMBL5288495_7vsi.pdbqt`, `v2d_pose_thymoquinone_7vsi.pdbqt`, `v2d_pose_dapagliflozin_7vsi.pdbqt`, `v2d_pose_analog_A_Me_CH2_Cl_7vsi.pdbqt`, `v2d_pose_analog_A_CF3_CH2_Cl_7vsi.pdbqt`, `v2d_pose_analog_A_Et_CH2_F_7vsi.pdbqt`, `v2d_pose_analog_A_Et_CH2_Me_7vsi.pdbqt`, `v2d_pose_analog_E_Et_Cl_S4_7vsi.pdbqt` (and the matching `v2e_pose_*_7vsi.pdbqt` files at exhaustiveness 16). Pose files: corrected-centre `v2c_pose_analog_A_Me_CH2_Cl_1us0.pdbqt`, `v2c_pose_analog_A_CF3_CH2_Cl_1us0.pdbqt`, `v2c_pose_analog_A_Et_CH2_F_1us0.pdbqt`, `v2c_pose_analog_A_Et_CH2_Me_1us0.pdbqt`, `v2c_pose_analog_E_Et_Cl_S4_1us0.pdbqt`; original-box `v2_pose_analog_A_Me_CH2_Cl_1us0.pdbqt`, `v2_pose_analog_A_CF3_CH2_Cl_1us0.pdbqt`, `v2_pose_analog_A_Et_CH2_F_1us0.pdbqt`, `v2_pose_analog_A_Et_CH2_Me_1us0.pdbqt`, `v2_pose_analog_E_Et_Cl_S4_1us0.pdbqt`, `v2_pose_chembl5288495_1us0.pdbqt`, `v2_pose_epalrestat_1us0.pdbqt`, `v2_redock_1us0_ldt.pdbqt`, `v2_redock_7vsi_7r3.pdbqt`. Original cross-docking: `cross_dock_results.json`, `sglt2_direct_results.json`, `controls_rerun_results.json` (same run directory). AutoDock Vina 1.2.7 throughout. This revision presents new computational data (listed above); no wet-lab data are included.